

This component is part of the data processing pipeline, primarily responsible for data normalization and integration. Data normalization is handled by parser classes (e.g., `ExcelDataParser`, `JSONDataParser`), while integration is performed by the `DataSourceAdapter`. The `DataListener` interface defines how data stream processing is started and stopped, enabling uniform handling of different data sources.

The parsers convert raw input data (e.g., CSV or JSON strings) into a standardized format by transforming it into a list of `IncomingDataPoint` objects, which contain the same core attributes as a patient record. During this process, invalid or incomplete records are filtered out to ensure data quality.

The `ExcelDataParser` processes CSV input by splitting the raw string into lines, validating each record, and converting valid entries into `IncomingDataPoint` objects. Similarly, the `JSONDataParser` handles JSON-formatted data, applying validation and transforming valid entries into the same standardized structure.

The `DataSourceAdapter` is responsible for integrating these normalized data points into the system. For each `IncomingDataPoint`, it creates a corresponding record and assigns it to the appropriate patient via the `IdentityManager`. If the patient does not yet exist, their creation is handled by another module, after which the new record is associated with that patient.